

Understanding User Mental Models Through App Sketches from Memory

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ABSTRACT

User-generated sketches serve as a powerful lens for understanding users' mental models of apps and services. Inspired by research on hand-drawn maps of cities, we conducted an exploratory study in which 123 participants drew the interface of a popular mobile app from memory. An analysis of the resulting sketches highlights that these types of artifacts can be used to understand which elements are most central to users' understanding of the app's core features and architecture—including non-existent features from other apps and services. We discuss how this work can be applied to the design of mobile apps and describe areas of future research.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies.

KEYWORDS

mental models, sketching, memory, user study, drawing, information architecture

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1 INTRODUCTION

Users' mental models are central to their beliefs about—and use of—complex systems, providing “predictive and explanatory power” [16] for understanding everyday use. In an industry setting where practitioners may develop an overly detailed understanding of their own applications, representations of their users' mental models can be a useful tool for empathy building and design inspiration, grounding designers and researchers in the often “messy, sloppy,

incomplete and indistinct” ways people actually understand the applications they use [16].

We were particularly interested in the use of sketching to elicit users' mental models, drawing inspiration from Stanley Milgram's 1970 “Psychological Maps of Paris” study [13] and follow-up research from Bentley et al. [2]. In his original study, Milgram asked 218 inhabitants of Paris to draw a map of their city from memory. The resulting sketches reflected individuals' personal experiences with Paris. For example, the study found that 84% of respondents included the Seine in their representations of Paris, while looking at the ways in which individuals mapped (and often distorted) the shape of the Seine could be traced back to their individual experience in the city. Milgram argues that this method allows researchers to understand how “reality is mirrored in the minds of its inhabitants,” and that sketching is a means through which to “render observable” an individual's mental model [13].

We present an exploratory study that extends the methodology of this previous sketching work—not to new types of maps but rather to modern app interfaces. Using the popular video-sharing app YouTube as a pilot case study, we asked 123 YouTube users to sketch the smartphone app interface from memory. Much like the Seine structured the maps of Paris in Milgram's work, we wanted to know which design elements of the YouTube app, both real and imagined, were most salient to users' mental models. In this paper, we present a preliminary analysis of the sketches drawn and discuss the viability and applications of using this approach to elicit user mental models.

2 PRIOR WORK

Users of apps and websites develop mental models that reflect which features are available on a given site [11, 18]. Nielsen [15] coined the term *Jakob's Law* to describe this phenomenon, arguing that user expectations around UI patterns and information architecture reflect users' *cumulative* experience, rather than a resetting of expectations across individual sites and apps. In our current study, we observed users' sketches often blurred the lines between apps and platforms as they reconstructed an app interface from memory.

Researchers have used a variety of methods and approaches to understand, elicit and represent users' mental models. Some have created heatmaps of general website layouts by having participants select cells on a grid to denote where they expected to locate certain features like navigation and search functions [3, 4, 19]. Participant eye-tracking is also commonly used to understand webpage mental



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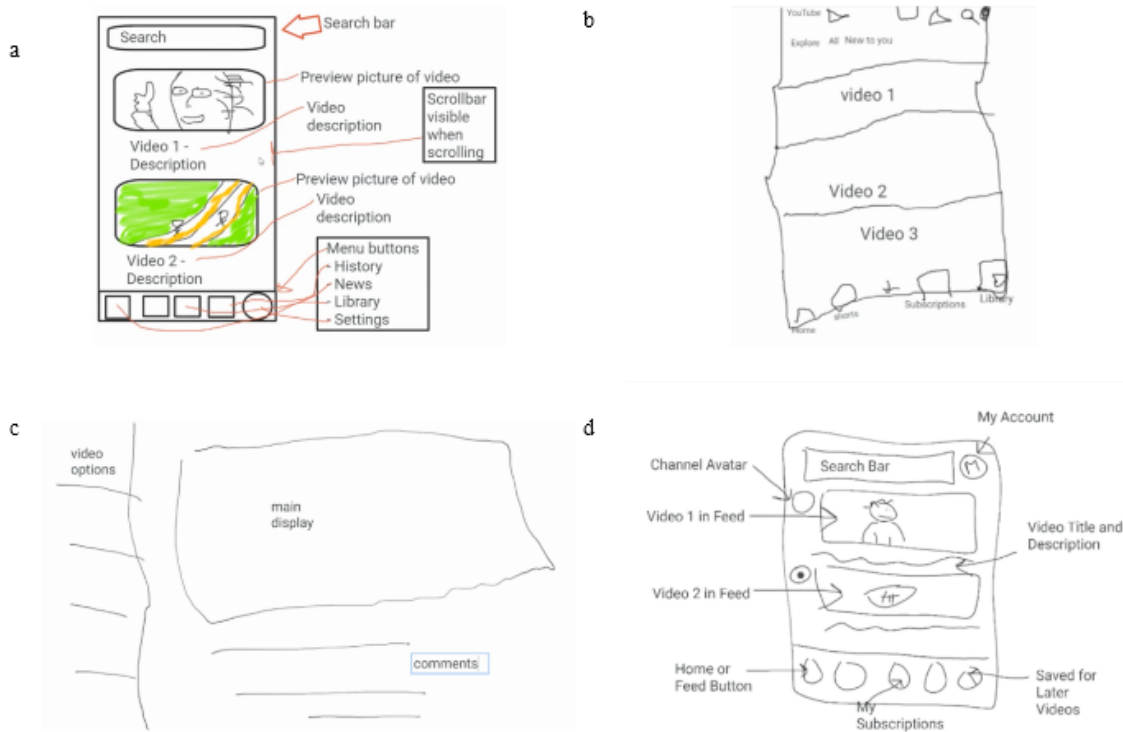


Figure 1: Example participant sketches from the 5-minute group [a,b] and 1-minute group [c,d]. While many of the features drawn exist in the YouTube app today, several features included do not, for example the History and News buttons in [a] and the search bar in [d].

models, observing where users focus their attention when completing certain types of tasks [5, 7, 8, 17].

Bentley et al. [2] replicated and updated Milgram’s psychological maps study in Chicago in the digital age, asking participants to create hand-drawn maps of “their Chicago” and correlating features of hand-drawn maps to demographic information and technology use. They found, for example, that the use of location-based check-in services was positively correlated with the number of neighborhoods participants sketched from memory.

More broadly, Sturdee and Lindley [20] documented sketching practices across HCI, such as sketching [6], storyboarding [10, 12] and diagramming [21], and have called for wider adoption of sketching within HCI practice. Indeed, participant sketches can be a powerful tool to help users collect and organize their beliefs and histories with technology (e.g., [14]). In this study, we add to this body of research by introducing an approach to eliciting app sketches from memory that make visible the often “messy” mental models with which users approach everyday app use.

3 METHOD

Data was collected using a remote, unmoderated usability testing platform that recorded users’ screens and captured user audio as

they completed the study. Participants completed their sketches using Jamboard (jamboard.google.com), a free, cross-platform whiteboarding tool that allows users to draw in a web browser using a mouse or touchscreen.

For the key study task, creating a sketch of the YouTube smartphone app from memory, participants were randomly assigned to a 1- or 5-minute time limit in which to complete their sketches. We added a time limit to this task for two reasons: first, we wanted to explore whether there was a difference in how users approached the sketching task based on sketching time; and second, we felt a time limit would better support the collection and analysis of user sketches at scale.

3.1 Participants

A total of 123 adult participants based in the United States completed the study using desktop computers with a keyboard and mouse. To qualify for the study, participants needed to own a smartphone with the YouTube app installed, indicate that the YouTube smartphone app was their primary way of accessing YouTube (versus other platforms such as desktop or tablet), and agree to use the Jamboard app as part of the study. Participants were diverse across age (ranging between 16-65 years, $M=33.6$, $SD=10.02$) and gender

(62 female, 60 male and 1 non-binary participant(s)). All participants completed an informed consent process prior to beginning the study, and were compensated for their time.

3.2 Procedure

Participants were recruited through a third-party participant panel. After completing a screener survey and informed consent, users received a general overview of study expectations and completed an initial set of familiarization tasks to understand the Jamboard platform and tools (e.g. pen tool, eraser tool, text tool). As a warm-up task, users were asked to sketch a drawing of a room in their house from memory, then add text labels indicating the various parts of the sketch.

Next, participants were assigned to a 1- or 5-minute and were then made aware of their core task: “Draw a sketch of the YouTube smartphone app from memory, including all the elements of the app that come to mind.” We intentionally left the task open-ended to give users more freedom to interpret what the “app” meant to them, and reminded participants that there was no right or wrong way to approach the task. Participants self-monitored their time using the clock provided within the usability testing tool. After completing the sketches, participants were asked to label all the elements in their sketch without a time limit, explain what they had sketched, and finally complete a short questionnaire. Screen recordings of participants’ study sessions were then collected for analysis.

3.3 Data Analysis

Participant sessions were screened for any deviations from the study instructions provided in the study task. User sketches that were not of the YouTube mobile app were removed from our corpus (e.g. desktop application, YouTube logo, irrelevant sketch), resulting in 123 sketches—59 1-minute sketches and 64 5-minute sketches—that were included for further analysis. For this preliminary pilot analysis, we specifically analyzed screenshots of participants’ completed sketches.

Sketches were analyzed using inductive thematic analysis [9]. This involves the development of “codes,” words or phrases to summarize an idea or theme from the data, which in our case lead to many codes about app elements due to the nature of the sketches. In this study, the codes developed were related to application elements observed in the sketches. The codebook was developed across two rounds of coding: first, two authors independently coded 20 sketches, then came together to consolidate and agree on codes and definitions. Three authors coded an additional set of 15 codes with this codebook, clarifying the definitions and expanding the codebook to 52 codes. The two authors completing the final sketch coding calibrated themselves by coding an identical set of 40 sketches and calculating their IRR to ensure their coding was consistent (Krippendorff’s $\alpha = 0.825$). After the final coding was completed, the researchers convened to discuss emerging themes, which are described below.

4 RESULTS

4.1 Overview

Key results from the analysis are shown in Table 1, which reflects the top-coded features in participant sketches. Coded elements included the orientation of the device sketched, the page or surface in the YouTube app represented, and which UI individual components were visible in the sketches. Participants sketched a wide range of features that reflected their mental models of the app.

4.2 Page and Phone Orientation

One of the key organizing structures for user sketches was the base page participants chose to represent. Though YouTube is an app focused on watching video content, the most commonly sketched app surface was the Home page (61.0% of sketches) where viewers browse video recommendations, rather than the Watch page (27.6% of sketches) for viewing. Accordingly, 75% of participants drew an app in portrait orientation, which is the default orientation for the Home page. Landscape orientation was most associated with the Watch page where videos are viewed, with 26.5% of Watch sketches reflecting a landscape view. Some participants also drew multiple screens to reflect their YouTube experience. For example, 2.7% of Home Feed sketches also included a separate Watch Page within the same sketch.

4.3 Most-Common and Niche UI Components

The most common structural UI elements drawn across all the sketches was a feed of video recommendations (85%) with a top navigation bar (83%) with a settings button (54.5%) and a YouTube logo (50%). Many users also included the bottom navigation tab (44%), which allows users to navigate across the various core pages of the app.

We also observed a variety of niche features included across the app sketches. For example, users labeled items such as the video upload date (1.6%), the closed captions button (1.6%), and the video quality settings button (0.8%). Some features more recently introduced to the app’s core UI, such as the prominent Shorts bottom navigation tab, introduced to the app last year, were only sketched by 7.3% of participants. Interestingly, some sketches, such as [d] in Figure 1, included an unlabeled placeholder for the Shorts tab showing that the participant remembered that an element belonged in that slot, but did not remember the specific element that belonged there.

4.4 Non-Existent UI Components

Participants also sketched features that do not currently exist in the app, including a hamburger menu (22%), a search bar in the top navigation (23%), a “history” bottom tab (2%), a “settings” bottom tab (2%) and in-player volume controls (0.8%).

4.5 Codes by Time Condition

Sketches in the 1- and 5-minute conditions received about the same number of codes during analysis. The 1-minute sketches received 10 median codes (min: 5, max 15) and the 5-minute sketches received 9 median codes (min: 2, max: 19). While the median number of features is nearly the same for 1-minute and 5-minute participants,

Table 1: Summary of Top-Coded Design Features in YouTube App Sketches*

Category	Item	All Participants (n=123)	5-min Condition (n=64)	1-min Condition (n=59)	Home Feed Drawings, (n=75)	Watch Page Drawings, (n=34)
Phone Orientation	Portrait	74.8%	78.1%	71.2%	84.0%	64.7%
	Unknown	17.1%	15.6%	18.6%	14.7%	8.8%
	Landscape	8.1%	6.3%	10.2%	1.3%	26.5%
App Page Drawn	Home Feed	61.0%	59.4%	62.7%	100.0%	5.9%
	Watch Page	27.6%	29.7%	25.4%	2.7%	100.0%
	Unknown Page	10.6%	10.9%	10.2%	0.0%	0.0%
Key UI Elements	Video Recs. Feed	85.4%	92.2%	78.0%	98.7%	67.6%
	Top Navigation	82.9%	87.5%	78.0%	94.7%	58.8%
	Bottom Navigation	43.9%	43.8%	44.1%	60.0%	8.8%
Top Nav. Elements	Video Player	26.0%	25.0%	27.1%	2.7%	88.2%
	Video Filters	12.2%	17.2%	6.8%	12.0%	5.9%
	Video Content Shelf	11.4%	12.5%	10.2%	14.7%	8.8%
	Account / Settings	54.5%	57.8%	50.8%	68.0%	32.4%
	YouTube Logo	49.6%	59.4%	39.0%	60.0%	32.4%
	Search Icon	33.3%	37.5%	28.8%	44.0%	11.8%
	Search Bar	22.8%	26.6%	18.6%	30.7%	14.7%
Video Feed Elements	Hamburger Menu	22.0%	18.8%	25.4%	30.7%	11.8%
	Video Thumbnail	37.4%	40.6%	33.9%	44.0%	17.6%
	Image					
Video Player Elements	Generic Video	18.7%	20.3%	16.9%	25.3%	5.9%
	Metadata					
	Video Title	13.8%	14.1%	13.6%	18.7%	8.8%
Watch Page Elements	Play Button	14.6%	17.2%	11.9%	4.0%	41.2%
Bottom Nav. Elements	Comments	12.2%	14.1%	10.2%	1.3%	44.1%
Bottom Nav. Elements	Home Tab	28.5%	34.4%	22.0%	40.0%	5.9%
	Subscriptions Tab	23.6%	31.3%	15.3%	32.0%	5.9%
	Library Tab	22.8%	23.4%	22.0%	30.7%	2.9%
	Create Tab	16.3%	15.6%	16.9%	22.7%	0.0%

* Design elements with less than 10% agreement omitted for space. Bolded cells represent codes with $\geq 50\%$ agreement.

the 5-minute cohort's sketches generated more codes than the 1-minute cohort, and as such, we observed a wider diversity of coded features across the 5-minute corpus.

5 DISCUSSION

5.1 Commonly Drawn Features and App Information Architecture

Looking at the most common features and structures in the sketches, a vague representation of the app begins to emerge when considering the most agreed upon features and structures: a video feed with a top navigation bar and YouTube logo. We consider these elements to reflect a kind of visual marker variable [1]—a single sketched UI component that signifies a more complex system in the users' mental model. As we investigated specific sections of the app, other marker variables emerged, like the video player on the Watch page (88% of Watch sketches) and the account/settings icon on the Home feed (68% of Home sketches).

These elements are not only what YouTube users consider most when thinking of the app, but they also are indicative of the key elements of the app's information architecture. Using the sketching method provides a novel and more visual way to determine how users think about the app, and its structures, and can also highlight areas of the app where users may experience higher levels of change aversion and frustration if altered.

We believe that non-existent features in participant sketches are also marker variables informed by users' collective experiences across apps as Nielsen describes in Jakob's Law [15]: users spend time across a variety of apps, observe common patterns, and develop expectations and intuitions about how apps should work. For example, users sketching a hamburger menu or a search bar—neither of which exist in the current YouTube app—likely represents a blurring of design elements across platforms. (Of note, the YouTube desktop site features both a hamburger menu and a search bar, and popular apps such as Instagram and Facebook also include hamburger menus.) From an information architecture perspective, this type of

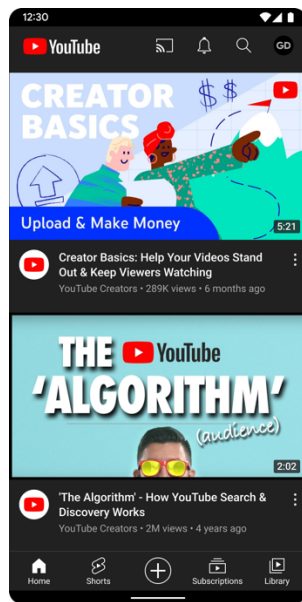


Figure 2: An example YouTube app Home page interface as it appeared in January 2022. Image courtesy of the authors.

analysis makes visible the types of structural elements users have expectations around given current visual design trends.

5.2 Differences in Sketch Time

Though we had originally hypothesized that participants given only one minute to sketch would draw fewer UI features per drawing, we found little difference between the 5-minute and 1-minute conditions by the total number of features represented. Instead, we observed that users across the five-minute condition drew a wider diversity of features. For example, when drawing the features of the YouTube video player, the 5-minute corpus of sketches included 10 distinct features, while the 1-minute corpus sketches only included 3, with the latter corpus identifying only the most prominent video player features such as the play button and the video progress bar.

As reflected in the colorful example sketch in Figure 1, we also found that 5-minute participants tended to use the extra sketching time to add more vibrancy to their sketches, testing out additional features in Jamboard and adding details such as color and thumbnail image sketches that helped better illustrate their mental models of the app. Given our interest in using sketches for empathy building in an industry setting, the 5-minute sketches with more diversity in detail and creativity are arguably more valuable to product teams since they offer an additional layer of detail regarding the design elements at the center of users' mental models.

6 CONCLUSION

In this study, we extended prior research on sketching by introducing an exploratory approach to eliciting app interface sketches from memory to understand user mental models. The data suggests that this method can provide a high-level understanding of which app elements are most salient among users, and that time

constraints and design conventions across devices and platforms can influence users' memory-elicitation. Overall, the study findings have important implications in terms of app redesigns (e.g., which features in the app are core to users' mental models and thus may be more susceptible to change aversion). We believe this is also a promising approach to more rapidly collect and explore data about users' mental models at scale.

Future work should further explore or control the relationships between participant demographics, behavioral data and sketches—for example participant age, frequency of smartphone use, or the prior experience or frequency of use with apps across the same genre or category. Additionally, future analysis could explore aspects of the sketches including the size, relative positions and naming of the elements. Research practitioners should explore the use of similar user app sketches as tools for empathy building in an industry setting. Given the digital nature of these sketches, participant fluency using digital sketching or whiteboard tools using a touchscreen, mouse, or keyboard should also be taken into consideration.

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